

Puzzle

- Many languages allow for a single morpheme to be interpreted as additive ('also') or scalar ('even'), such as Marathi *-hī* (1).
- (1) *Rām-ne-hī pustak vach-l-e.*
Ram-ERG.SG-**ADD** book read-PFV-N.SG
'Ram also read the book.' or 'Even Ram read the book.'
- hī** can also give rise to a maximal reading with a numeral $n \geq 2$.
- (2) *Tīn-hī mulla-nnī ek āmbā khal-l-a.*
three-**ADD** boy-ERG.PL one mango eat-PFV-M.SG
'(All) the three boys ate one mango.'
- The same alternation occurs in Magahi, Banjara, Ewe, Kazakh, Balkar (Bylinina et al. 2020), Turkish (Sağ-Parvardeh 2019: 183), Greek (Chatzikyriakidis et al. 2015: 178), Sanzhi Dargwa (Forker 2020: 136), Agul (Maisak 2015: 4), Laz (Lacroix 2009: 102), Tamil (Lehmann 1993: 152), Telugu, Malayalam (Hany Babu 2006: 13) and Amharic (Demeke & Meyer 2008: 621).

'Even' and domain maximality: Bylinina et al. (2020)

- Two presuppositions can be associated with the additive morpheme.
- (3) $\text{ADD}(\alpha) \rightsquigarrow \exists \phi \in \llbracket \alpha \rrbracket^f \setminus \llbracket \alpha \rrbracket^o [\phi \text{ true}]$
- (4) $\text{SCAL}(\alpha) \rightsquigarrow \forall \phi \in \llbracket \alpha \rrbracket^f \setminus \llbracket \alpha \rrbracket^o [\llbracket \alpha \rrbracket^o <_{\text{likely}} \phi]$
- Bylinina et al. propose the maximal numeral reading should be derived from the 'even' use, with focus alternatives in (5).
- (5) $\llbracket (2) \rrbracket^f = \{\lambda w. 1 \text{ boy ate one mango in } w, \lambda w. 2 \text{ boys ate one mango in } w, \dots\}$
- The lower numerals satisfy ADD (3), and if likelihood is based on entailment, SCAL (4) is satisfied only in contexts with exactly 3 boys.
 - However, defining an 'even' use with entailment rather than likelihood is non-standard.
 - And likelihood cannot be used because not every predicate is such that it's more likely for a smaller number of individuals to satisfy the predicate.
 - Another issue is that the maximal interpretation is compatible with collective predicates (6).
- (6) *Don-hī mul-e mitra ah-et.*
two-**add** boy-PL friend COP.PRS-3PL
'The two boys are friends.'
- So, the additive morpheme cannot be taking the asserted proposition as its argument, since additivity will not be satisfied (7).
 - This would mean we lose the reasoning for restricting maximality to $n \geq 2$.
- (7) *The two boys are friends. $\nrightarrow$ #The one boy is friends.*
- And, some languages have a maximal use but not an 'even' use (Tamil, Malayalam; Iyer 2017).

The 'also'/'even' alternation: Xiang (2026)

- Xiang (2026) works on the Mandarin additive/scalar *ye* in the framework of Heim (1982, 1983).
- (8) For any context c and sentences ϕ and ψ :
- $$c + ye_{\psi}(\phi) = \begin{cases} c + \phi, & \text{if } c + \phi = c + \text{not-ONLY}_{\phi}^{(L)}(\psi) \\ \text{undefined}, & \text{otherwise} \end{cases}$$
- Additivity uses logical strength ONLY (9), while scalarity uses likelihood ONLY (10).
- (9) $c + \text{ONLY}_{\phi}(\psi) = \begin{cases} c + \text{not-}\phi & \text{if } \psi \in \text{F-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi \nRightarrow \phi \\ c & \text{if } \psi \in \text{F-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi \Rightarrow \phi \\ \text{undefined} & \text{otherwise} \end{cases}$
- (10) $c + \text{ONLY}_{\phi}^L(\psi) = \begin{cases} c + \text{not-}\phi & \text{if } \psi \in \text{F-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi >_{c\text{-likely}} \phi \\ c & \text{if } \psi \in \text{F-ALT}(\phi), c \subseteq \llbracket \psi \rrbracket, \text{ and } \psi \leq_{c\text{-likely}} \phi \\ \text{undefined} & \text{otherwise} \end{cases}$

Derving maximality from 'also'

- Goal: derive the meaning of **-hī** in (2):
- directly from its additive use
 - in a way that explains why only $n \geq 2$ can be used
 - in a way consistent with approaches to the 'also'/'even' alternation
 - without running into an issue with collective predicates
- Rather than **-hī** operating directly on the asserted proposition in (2) it must operate on something like (11) with focus alternatives (12).
- (11) $\llbracket \phi_3 \rrbracket = \text{There are 3 boys.}$
- (12) $\llbracket \phi_3 \rrbracket^f = \{\phi_1, \phi_2, \phi_4, \phi_5, \dots\}$
- This avoids the issue with collective predicates and additivity (6), since now we have (13).
- (13) *There are two boys. $\rightarrow$ There is one boy.*
- In the recursive exhaustification approach, maximality comes from (18a).
- (14) $O_{ALT}(\phi_3) = \phi_3 \wedge \forall p \in \text{ALT}[p \rightarrow \phi_3 \subseteq p]$
- This says that the only true focus alternatives are those that are entailed by ϕ_3 .
 - So, $\phi_4, \phi_5, \dots$ must all be false, giving us the maximality effect.
 - In the context-change potential approach, we can use the definedness condition on logical strength ONLY.
- (15) $c + ye_{\psi}^{\text{ALSO}}(\phi)$ is defined only if
- $\psi \in \text{F-ALT}(\phi)$ focus-sensitivity
 - $c \subseteq \llbracket \psi \rrbracket$ and $\psi \nRightarrow \phi$ additivity
- This says that the focus alternatives must be entailed by the context and cannot entail ϕ_3 .
 - Once again, this excludes the higher alternatives $\phi_4, \phi_5, \dots$
 - And, additivity still requires there to be a true focus alternative.
 - Within both these approaches, this derives the maximality effect without using the machinery needed for scalarity.

The 'also'/'even' alternation: Kirby (2023)

- Kirby (2023) uses the following exhaustifiers from (Chierchia 2013: 31,148)
- (16) $O_{ALT}(\phi) = \phi \wedge \forall p \in \text{ALT}[p \rightarrow \phi \subseteq p]$, where $\subseteq$ means 'entails'
- (17) $E_{ALT}(\phi) = \phi \wedge \forall p \in \text{ALT}[\phi <_{\mu} p]$, where $\phi <_{\mu} p$ says that ϕ is less likely than p with respect to some contextually relevant probability measure μ
- Additivity comes from recursive exhaustification (18a)-(18b).
 - Scalarity comes from proceeding to E_{ALT} (18c).
- (18) **Optimal Fit.** In exhaustifying ϕ : (18a)>(18b)>(18c)
- Use O unless $O_{ALT}(\phi)$ is trivial (contradictory or vacuous).
 - Repeat exhaustifying with O.
 - If there is a salient probability metric $\mathcal{L}$, use E_{ALT} .

Open Questions

- Although these alternations are quite common cross-linguistically, not every particle in every language always has every possible meaning associated with that kind of particle.
- For example, Japanese *-mo* and Korean *-do* can alternate between 'also' and 'even', but do not have a maximal reading with numerals.
- Based on this paper, Kirby (2023), and Szabolcsi (2015), (19) gives a list of possible uses of additive morphemes and their instantiations in a few languages.

(19)

Language	Magahi	Marathi	BCS	Hungarian	Japanese
Role					
everyone (GQ), $\forall$ anyone, FCI anyone, NPI	X kaun- o kaun- o	X koṇi- hī koṇi- hī	X X i -(t)ko, ni -(t)ko	X akár-ki is akár-ki is , vala-ki is , sen -ki	daré- mo dare-de- mo dare- mo
both X and Y neither X nor Y	?X- o Y- o ?X- o Y- o	? ?	i X i Y ni X ni Y	X is Y is X sem Y sem , sem X sem Y	X- mo Y- mo X- mo Y- mo
X too even X	X- o X- o	X- hī X- hī	i X (čak) i X	X is még X is	X- mo X- mo

- Establishing a way to derive all these possible uses in a uniform way is an interesting puzzle, but so is figuring out how to constrain the uses of an additive morpheme in a singular language.
- There is another open question with respect to the distributivity of maximal numeral readings.
- Sağ-Parvardeh (2019: 172) reports for Turkish that *dA* has a maximal numeral reading, but it must be distributive.

(20) *İki çocuk da bir şişe süt içti.*
two boy ADD one bottle milk drink-PST

'The two boys drank one bottle of milk.' (distributive - two bottles)